Project

End-to-End E-commerce Intelligence System: Building a Customer 360 Analytics Framework

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1. Business Problem Statement

In modern e-commerce ecosystems, data is generated across multiple independent systems such as customer management, order processing, payments, product catalogs, seller networks, and customer feedback platforms. These datasets are typically fragmented and stored in separate tables, making it difficult to extract unified insights.

The objective of this project is to simulate a real-world data analytics scenario where you are required to:

- Integrate multiple data sources into a unified analytical dataset
- Construct a **Customer 360 View**
- Analyze customer behavior, revenue patterns, and operational performance
- Identify key drivers of business growth and customer satisfaction
- Generate actionable, data-driven business recommendations

This project reflects how data analysts and data scientists work in real organizations, where raw data must be transformed into meaningful insights before applying machine learning models.

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2. Dataset Overview

You are provided with a multi-table e-commerce dataset consisting of the following files (renamed for simplicity):

Core Tables

1. **customers.csv**
   Contains customer demographic and location information
2. **orders.csv**
   Central table containing order lifecycle details (purchase, delivery, timestamps)
3. **order_items.csv**
   Contains product-level details for each order

4. **payments.csv**
   Payment information including type and value

5. **reviews.csv**
   Customer feedback and review scores

Supporting Tables

6. **products.csv**
   Product details and categories

7. **sellers.csv**
   Seller-level information

8. **geolocation.csv**
   Geographic information (optional for advanced analysis)

9. **category_translation.csv**
   Mapping of product categories to English names

Important Note

These datasets are relational and must be joined using appropriate keys such as:

- order_id
- customer_id
- product_id
- seller_id

The **orders table acts as the central fact table** connecting most datasets.

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3. Project Objectives

By completing this project, students will:

- Develop the ability to work with multi-table datasets
- Learn data cleaning and preprocessing techniques
- Perform data integration (joins/merges)

- Conduct exploratory data analysis (EDA)
- Apply feature engineering techniques
- Build strong data visualization skills
- Generate business insights from raw data

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4. Project Execution Guidelines

Step 1: Data Loading and Initial Exploration

- Load all datasets using Pandas
- Inspect structure using .head(), .info(), .describe()
- Identify primary and foreign keys
- Understand relationships between tables

Step 2: Data Cleaning and Preprocessing

- Handle missing values appropriately
- Remove duplicate records
- Convert date columns to datetime format
- Validate data types and ranges
- Standardize column names if required

Step 3: Data Integration (Critical Component)

You must construct a **Master Dataset** by merging multiple tables.

Recommended sequence:

1. orders + customers
2. orders + order_items
3. order_items + products

4. orders + payments
5. orders + reviews
6. order_items + sellers
7. products + category_translation

Final output:
A **single consolidated dataset** representing a unified business view

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Step 4: Feature Engineering

Create meaningful features such as:

- Total order value (aggregated from order_items or payments)
- Delivery time (order purchase to delivery date)
- Number of items per order
- Customer purchase frequency
- Customer lifetime value (basic approximation)
- Average order value per customer

Step 5: Exploratory Data Analysis (EDA)

Perform structured analysis across the following dimensions:

Customer Analysis

- New vs repeat customers
- High-value vs low-value customers
- Geographic distribution of customers

Revenue and Order Analysis

- Monthly revenue trends

- Order volume trends
- Peak sales periods

Product Analysis

- Top-selling product categories
- Revenue contribution by category
- Product demand distribution

Seller Analysis

- Top-performing sellers
- Seller contribution to revenue
- Seller distribution

Review and Satisfaction Analysis

- Distribution of review scores
- Relationship between delivery time and ratings
- Identification of dissatisfaction patterns

Step 6: Data Visualization

Use Matplotlib and Seaborn to create:

- Time series plots (sales trends)
- Bar charts (category performance)
- Histograms (distribution analysis)
- Box plots (outlier detection)
- Heatmaps (correlation analysis)

All visualizations must be clearly labeled and interpretable.

Step 7: Business Insights and Recommendations

You must derive clear and actionable insights:

- Identify top revenue-driving factors
- Highlight customer behavior patterns
- Evaluate operational inefficiencies
- Provide strategic recommendations

Insights must be supported by data and visual evidence.

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5. Deliverables

Each student must submit:

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1. Jupyter Notebook (.ipynb)

- Clean, well-structured code
- Proper comments and explanations
- Logical step-by-step flow

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2. Project Report (PDF/DOC)

Must include:

- Project Title
- Business Problem Statement
- Dataset Description
- Methodology
- Key Findings

- Business Insights
- Recommendations

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3. Visual Outputs

- All key charts and graphs
- Properly labeled and readable

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4. GitHub Repository (Recommended)

Students should upload:

- Notebook
- README.md
- Dataset (sample or link)

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6. Submission Guidelines

- Submit all files in a single folder
- Folder naming convention:
  Name_Ecommerce_Intelligence_Project
- Ensure:
  - Code runs without errors
  - Proper structure is maintained
  - Files are clearly named

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7. Evaluation Criteria

Students will be assessed based on:

- Data preprocessing and cleaning quality
- Accuracy of data merging

- Depth of analysis
- Quality and clarity of visualizations
- Business understanding
- Insight generation and recommendations
- Code readability and structure

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8. Academic Integrity

- Original work is expected
- Do not copy solutions from external sources
- Proper understanding is required for evaluation